SIM1003 Calculus II
Tutorial 5

All even-numbered questions in the following exercises

Exercise 9.4:

Determining Convergence or Divergence

Which of the series in Exercises 17–56 converge, and which diverge? Use any method, and give reasons for your answers.

17. $\sum_{n=1}^{\infty} \frac{1}{2\sqrt{n} + \sqrt[3]{n}}$

18. $\sum_{n=1}^{\infty} \frac{3}{n + \sqrt{n}}$

19. $\sum_{n=1}^{\infty} \frac{\sin^2 n}{2^n}$

20. $\sum_{n=1}^{\infty} \frac{1 + \cos n}{n^2}$

21. $\sum_{n=1}^{\infty} \frac{2n}{3n - 1}$

22. $\sum_{n=1}^{\infty} \frac{n + 1}{n^2\sqrt{n}}$

23. $\sum_{n=1}^{\infty} \frac{10n + 1}{n(n + 1)(n + 2)}$

24. $\sum_{n=3}^{\infty} \frac{5n^3 - 3n}{n^2(n - 2)(n^2 + 5)}$

25. $\sum_{n=1}^{\infty} \left(\frac{n}{3n + 1}\right)^n$

26. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3 + 2}}$

27. $\sum_{n=3}^{\infty} \frac{1}{\ln(\ln n)}$

28. $\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^3}$

29. $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n} \ln n}$

30. $\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^{3/2}}$

31. $\sum_{n=1}^{\infty} \frac{1}{1 + \ln n}$

32. $\sum_{n=2}^{\infty} \frac{\ln(n + 1)}{n + 1}$

33. $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{n^2 - 1}}$

34. $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2 + 1}$

35. $\sum_{n=1}^{\infty} \frac{1 - n}{n2^n}$

36. $\sum_{n=1}^{\infty} \frac{n + 2^n}{n^2 2^n}$

37. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{3^{n-1}+1}$

38. $\displaystyle\sum_{n=1}^{\infty} \frac{3^{n-1}+1}{3^n}$

39. $\displaystyle\sum_{n=1}^{\infty} \frac{n+1}{n^2+3n} \cdot \frac{1}{5n}$

40. $\displaystyle\sum_{n=1}^{\infty} \frac{2^n+3^n}{3^n+4^n}$

41. $\displaystyle\sum_{n=1}^{\infty} \frac{2^n-n}{n2^n}$

42. $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n}{\sqrt{n}\, e^n}$

43. $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n!}$

(*Hint:* First show that $(1/n!) \le (1/n(n-1))$ for $n \ge 2$.)

44. $\displaystyle\sum_{n=1}^{\infty} \frac{(n-1)!}{(n+2)!}$

45. $\displaystyle\sum_{n=1}^{\infty} \sin\frac{1}{n}$

46. $\displaystyle\sum_{n=1}^{\infty} \tan\frac{1}{n}$

47. $\displaystyle\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^{1.1}}$

48. $\displaystyle\sum_{n=1}^{\infty} \frac{\operatorname{arcsec} n}{n^{1.3}}$

49. $\displaystyle\sum_{n=1}^{\infty} \frac{\coth n}{n^2}$

50. $\displaystyle\sum_{n=1}^{\infty} \frac{\tanh n}{n^2}$

51. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n\sqrt[n]{n}}$

52. $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt[n]{n}}{n^2}$

53. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{1+2+3+\cdots+n}$

54. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{1+2^2+3^2+\cdots+n^2}$

55. $\displaystyle\sum_{n=2}^{\infty} \frac{n}{(\ln n)^2}$

56. $\displaystyle\sum_{n=2}^{\infty} \frac{(\ln n)^2}{n}$

Exercise 9.5:

In Exercises 1–16, use the Ratio Test or the Ratio Test to determine whether each converges absolutely or diverges.

Exercises 1–8: Using the Ratio Test

1. $\sum_{n=1}^{\infty} \frac{2^n}{n!}$

2. $\sum_{n=1}^{\infty} (-1)^n \frac{n+2}{3^n}$

3. $\sum_{n=1}^{\infty} \frac{(n-1)!}{(n+1)^2}$

4. $\sum_{n=1}^{\infty} \frac{2^{n+1}}{n3^{n-1}}$

5. $\sum_{n=1}^{\infty} \frac{n^4}{(-4)^n}$

6. $\sum_{n=2}^{\infty} \frac{3^{n+2}}{\ln n}$

7. $\sum_{n=1}^{\infty} (-1)^n \frac{n^2(n+2)!}{n!3^{2n}}$

8. $\sum_{n=1}^{\infty} \frac{n5^n}{(2n+3)\ln(n+1)}$

Exercises 9 – 16: Using the Root Test

9. $\sum_{n=1}^{\infty} \frac{7}{(2n+5)^n}$

10. $\sum_{n=1}^{\infty} \frac{4^n}{(3n)^n}$

11. $\sum_{n=1}^{\infty} \left(\frac{4n+3}{3n-5}\right)^n$

12. $\sum_{n=1}^{\infty} \left(-\ln\left(e^2 + \frac{1}{n}\right)\right)^{n+1}$

13. $\sum_{n=1}^{\infty} \frac{-8}{\left(3+(1/n)\right)^{2n}}$

14. $\sum_{n=1}^{\infty} \sin^n\left(\frac{1}{\sqrt{n}}\right)$

15. $\sum_{n=1}^{\infty} (-1)^n \left(1 - \frac{1}{n}\right)^{n^2}$

(*Hint:* $\lim_{n\to\infty}(1 + x/n)^n = e^x$)

16. $\sum_{n=2}^{\infty} \frac{(-1)^n}{n^{1+n}}$

Determining Convergence or Divergence

In Exercises 17 – 54, use any method to determine whether the series converges or diverges. Give reasons for your answer.

17. $\sum_{n=1}^{\infty} \frac{n^{\sqrt{2}}}{2^n}$

18. $\sum_{n=1}^{\infty} (-1)^n n^2 e^{-n}$

19. $\sum_{n=1}^{\infty} n!(-e)^{-n}$

20. $\sum_{n=1}^{\infty} \frac{n!}{10^n}$

21. $\sum_{n=1}^{\infty} \frac{n^{10}}{10^n}$

22. $\sum_{n=1}^{\infty} \left(\frac{n-2}{n}\right)^n$

23. $\sum_{n=1}^{\infty} \frac{2+(-1)^n}{1.25^n}$

24. $\sum_{n=1}^{\infty} \frac{(-2)^n}{3^n}$

25. $\sum_{n=1}^{\infty} (-1)^n \left(1 - \frac{3}{n}\right)^n$

26. $\displaystyle\sum_{n=1}^{\infty}\left(1-\frac{1}{3n}\right)^n$

27. $\displaystyle\sum_{n=1}^{\infty}\frac{\ln n}{n^3}$

28. $\displaystyle\sum_{n=1}^{\infty}\frac{(-\ln n)^n}{n^n}$

29. $\displaystyle\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n^2}\right)$

30. $\displaystyle\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n^2}\right)^n$

31. $\displaystyle\sum_{n=1}^{\infty}\frac{e^n}{n^e}$

32. $\displaystyle\sum_{n=1}^{\infty}\frac{n\ln n}{(-2)^n}$

33. $\displaystyle\sum_{n=1}^{\infty}\frac{(n+1)(n+2)}{n!}$

34. $\displaystyle\sum_{n=1}^{\infty}e^{-n}\left(n^3\right)$

35. $\displaystyle\sum_{n=1}^{\infty}\frac{(n+3)!}{3!n!3^n}$

36. $\displaystyle\sum_{n=1}^{\infty}\frac{n2^n(n+1)!}{3^n n!}$

37. $\displaystyle\sum_{n=1}^{\infty}\frac{n!}{(2n+1)!}$

38. $\displaystyle\sum_{n=1}^{\infty}\frac{n!}{(-n)^n}$

39. $\displaystyle\sum_{n=2}^{\infty}\frac{-n}{(\ln n)^n}$

40. $\displaystyle\sum_{n=2}^{\infty}\frac{n}{(\ln n)^{(n/2)}}$

41. $\displaystyle\sum_{n=1}^{\infty}\frac{n!\ln n}{n(n+2)!}$

42. $\displaystyle\sum_{n=1}^{\infty}\frac{(-3)^n}{n^3 2^n}$

43. $\displaystyle\sum_{n=1}^{\infty}\frac{(n!)^2}{(2n)!}$

44. $\displaystyle\sum_{n=1}^{\infty}\frac{(2n+3)(2^n+3)}{3^n+2}$

45. $\displaystyle\sum_{n=3}^{\infty}\frac{2^n}{n^2}$

46. $\displaystyle\sum_{n=3}^{\infty}\frac{2^{n^2}}{n^{2^n}}$

47. $\displaystyle\sum_{n=1}^{\infty}\frac{2^n n!n!}{(2n)!}$

48. $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n(3n)!}{n!(n+1)!(n+2)!}$

49. $\displaystyle\sum_{n=1}^{\infty}\frac{(n!)^n}{(n^n)^2}$

50. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{(n!)^n}{n^{(n^2)}}$

51. $\displaystyle\sum_{n=1}^{\infty}\frac{n^n}{2^{(n^2)}}$

52. $\displaystyle\sum_{n=1}^{\infty}\frac{n^n}{(2^n)^2}$

53. $\displaystyle\sum_{n=1}^{\infty}\frac{1\cdot 3\cdot\cdots\cdot(2n-1)}{4^n 2^n n!}$

54. $\displaystyle\sum_{n=1}^{\infty}\frac{1\cdot 3\cdot\cdots\cdot(2n-1)}{[2\cdot 4\cdot\cdots\cdot(2n)](3^n+1)}$

Exercise 9.6:

Convergence of Alternating Series

In Exercises 1–14, determine whether the alternating series converges or diverges.

1. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt{n}}$
2. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^{3/2}}$
3. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n3^n}$
4. $\sum_{n=2}^{\infty} (-1)^{n} \frac{4}{(\ln n)^2}$
5. $\sum_{n=1}^{\infty} (-1)^{n} \frac{n}{n^2+1}$
6. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2+5}{n^2+4}$
7. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2^n}{n^2}$
8. $\sum_{n=1}^{\infty} (-1)^{n} \frac{10^n}{(n+1)!}$
9. $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{n}{10}\right)^n$
10. $\sum_{n=2}^{\infty} (-1)^{n+1} \frac{1}{\ln n}$
11. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\ln n}{n}$
12. $\sum_{n=1}^{\infty} (-1)^{n} \ln\left(1+\frac{1}{n}\right)$
13. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sqrt{n}+1}{n+1}$
14. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3\sqrt{n+1}}{\sqrt{n}+1}$

Absolute and Conditional Convergence

Which of the series in Exercises 15–48 converge absolutely, which converge conditionally, and which diverge? Give reasons for your answers.

15. $\sum_{n=1}^{\infty} (-1)^{n+1} (0.1)^n$
16. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{(0.1)^n}{n}$
17. $\sum_{n=1}^{\infty} (-1)^{n} \frac{1}{\sqrt{n}}$
18. $\sum_{n=1}^{\infty} \frac{(-1)^n}{1+\sqrt{n}}$
19. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{n^3+1}$
20. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n!}{2^n}$
21. $\sum_{n=1}^{\infty} (-1)^{n} \frac{1}{n+3}$
22. $\sum_{n=1}^{\infty} (-1)^{n} \frac{\sin n}{n^2}$
23. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3+n}{5+n}$
24. $\sum_{n=1}^{\infty} \frac{(-2)^{n+1}}{n+5^n}$
25. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1+n}{n^2}$
26. $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\sqrt[n]{10}\right)$
27. $\sum_{n=1}^{\infty} (-1)^{n} n^2 (2/3)^n$
28. $\sum_{n=2}^{\infty} (-1)^{n+1} \frac{1}{n \ln n}$

29. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{\arctan n}{n^2+1}$

30. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{\ln n}{n-\ln n}$

31. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{n}{n+1}$

32. $\displaystyle\sum_{n=1}^{\infty}(-5)^{-n}$

33. $\displaystyle\sum_{n=1}^{\infty}\frac{(-100)^n}{n!}$

34. $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^2+2n+1}$

35. $\displaystyle\sum_{n=1}^{\infty}\frac{\cos n\pi}{n\sqrt{n}}$

36. $\displaystyle\sum_{n=1}^{\infty}\frac{\cos n\pi}{n}$

37. $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n(n+1)^n}{(2n)^n}$

38. $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n+1}(n!)^2}{(2n)!}$

39. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{(2n)!}{2^n n!n}$

40. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{(n!)^2 3^n}{(2n+1)!}$

41. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\left(\sqrt{n+1}-\sqrt{n}\right)$

42. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\left(\sqrt{n^2+n}-n\right)$

43. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\left(\sqrt{n+\sqrt{n}}-\sqrt{n}\right)$

44. $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n}{\sqrt{n}+\sqrt{n+1}}$

45. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\operatorname{sech} n$

46. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\operatorname{csch} n$

47. $\displaystyle\frac{1}{4}-\frac{1}{6}+\frac{1}{8}-\frac{1}{10}+\frac{1}{12}-\frac{1}{14}+\cdots$

48. $\displaystyle 1+\frac{1}{4}-\frac{1}{9}-\frac{1}{16}+\frac{1}{25}+\frac{1}{36}-\frac{1}{49}-\frac{1}{64}+\cdots$

Error Estimation

In Exercises 49–52, estimate the magnitude of the error involved in using the sum of the first four terms to approximate the sum of the entire series.

49. $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}$

50. $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{10^n}$

51. $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\frac{(0.01)^n}{n}$ As you will see in Section10.7, the sum is ln(1.01).

52. $\displaystyle\frac{1}{1+t}=\sum_{n=0}^{\infty}(-1)^n t^n, \quad 0<t<1$

In Exercises 53–56, determine how many terms should be used to estimate the sum of the entire series with an error of less than 0.001.

53. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{1}{n^2+3}$

54. $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n}{n^2+1}$

55. $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{\left(n+3\sqrt{n}\right)^3}$

56. $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{1}{\ln(\ln(n+2))}$